CS24305 — THEORY (3.0 CR) MODULE V NOTES BIT MESRA NEP SCHEME

Module 5: Transport Layer & Security

TCP 3–Way Handshake, TCP AIMD Congestion Control, DNS Hierarchy, HTTP Evolution & RSA Public-Key Math

Module V: Transport Layer, Application Services & Network Security

1. Transport Layer Port Multiplexing & Sockets Interface (TCP vs. UDP)
2. TCP Segment Anatomy: Sequence Numbers, Acknowledgments & Control Flags
3. TCP 3–Way Handshake Connection Establishment & 4–Way FIN Teardown
4. TCP Flow Control (Receiver Window `rwnd` & Nagle's Algorithm)
5. TCP Congestion Control: Slow Start, Congestion Avoidance (AIMD), Fast Retransmit
6. TCP Fast Recovery & Cubic / BBR Modern Congestion Mechanisms
7. Domain Name System (DNS): Hierarchical Namespace, Root Servers & Records (A, AAAA, MX)
8. Hypertext Transfer Protocol Evolution: HTTP/1.1 vs. HTTP/2 vs. HTTP/3 (QUIC)
9. Email Architecture: SMTP, POP3, IMAP & MIME Extensions
10. Cryptographic Fundamentals: Symmetric (AES) vs. Asymmetric (RSA) Cryptography
11. RSA Public-Key Algorithm Mathematical Derivation & Worked Encryption Traces
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 1 – 3: TCP Protocol Architecture & 3–Way Handshake

TCP Control Flag	Bit Position	Operational Role in Connection State Machine
SYN	Synchronize	Initiates a connection; synchronizes initial sequence numbers (ISN). Consumes 1 sequence number.
ACK	Acknowledgment	Validates the 32-bit Acknowledgment Number field indicating next expected byte.
FIN	Finish	Sender has finished sending data and requests connection termination. Consumes 1 sequence number.
RST	Reset	Abruptly terminates an invalid or unauthorized connection request.
PSH	Push	Instructs receiver to deliver buffered data to application immediately without waiting for buffer to fill.
URG	Urgent	Validates the Urgent Pointer field for out-of-band high-priority data (e.g., `Ctrl+C`).

🏛 Step-by-Step Solved Problem: TCP 3-Way Handshake Connection Sequence

1. **Client $\rightarrow$ Server (SYN):** Client chooses random initial sequence number x:

$$\text{Segment 1: } [\mathbf{SYN} = 1,\ \mathbf{Seq} = x]$$

2. **Server $\rightarrow$ Client (SYN-ACK):** Server acknowledges client's ISN ($x + 1$) and chooses its own ISN y:

$$\text{Segment 2: } [\mathbf{SYN} = 1,\ \mathbf{ACK} = 1,\ \mathbf{Seq} = y,\ \mathbf{Ack} = x + 1]$$

3. **Client $\rightarrow$ Server (ACK):** Client acknowledges server's ISN ($y + 1$):

$$\text{Segment 3: } [\mathbf{ACK} = 1,\ \mathbf{Seq} = x + 1,\ \mathbf{Ack} = y + 1]$$

Connection established in ESTABLISHED state.

Topic 5 & 6: TCP Congestion Control Mechanisms (AIMD)

The 4 Standard Phases of TCP Congestion Control (Tahoe / Reno):

1. **Slow Start:** Initial congestion window $\text{cwnd} = 1 \text{ MSS}$. Doubles every Round-Trip Time ($\text{cwnd} \leftarrow \text{cwnd} \times 2$) exponentially until $\text{cwnd} \geq \text{ssthresh}$ (Slow Start Threshold).
2. **Congestion Avoidance (Additive Increase):** Increases linearly by 1 MSS per RTT ($\text{cwnd} \leftarrow \text{cwnd} + 1$).
3. **Triple Duplicate ACK Event (Fast Retransmit & Recovery):** - $\text{ssthresh} \leftarrow \frac{\text{cwnd}}{2}$ - $\text{cwnd} \leftarrow \text{ssthresh} + 3 \text{ MSS}$ (TCP Reno skips Slow Start and continues linearly).
4. **Timeout Event (Severe Congestion):** - $\text{ssthresh} \leftarrow \frac{\text{cwnd}}{2}$ - $\text{cwnd} \leftarrow 1 \text{ MSS}$ (Drops back to Slow Start).

Topic 10 & 11: Network Security & RSA Public-Key Cryptography

RSA Algorithm Mathematical Formulations (Rivest, Shamir, Adleman, 1977):

1. Select two distinct large prime numbers p and q.
2. Compute modulus $n = p \times q$.
3. Compute Euler's Totient $\phi(n) = (p - 1)(q - 1)$.
4. Choose public exponent e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
5. Compute private decryption exponent d using Extended Euclidean Algorithm:

$$d \equiv e^{-1} \pmod{\phi(n)} \implies (d \times e) \pmod{\phi(n)} = 1$$

6. **Public Key:** $KU = \{e, n\}$, **Private Key:** $KR = \{d, n\}$.
7. **Encryption:** Ciphertext $C = M^e \pmod n$.
8. **Decryption:** Plaintext $M = C^d \pmod n$.

🏛 Step-by-Step Solved Problem: Complete RSA Encryption & Decryption Trace

Let primes $p = 3, q = 11$, and Plaintext message $M = 7$.

1. $n = 3 \times 11 = \mathbf{33}$.
2. $\phi(n) = (3 - 1)(11 - 1) = 2 \times 10 = \mathbf{20}$.
3. Choose $e = 7$ ($\gcd(7, 20) = 1$).
4. Find d: $(d \times 7) \pmod{20} = 1 \implies 7d = 21 \implies \mathbf{d = 3}$.
5. **Encryption:** $C = 7^7 \pmod{33} = 823543 \pmod{33} = \mathbf{4}$.
6. **Decryption:** $M = 4^3 \pmod{33} = 64 \pmod{33} = \mathbf{7}$. (Original plaintext recovered!).

🏛️ Top BIT Mesra Exam Questions & Answers (Module V)

Q1. Compare HTTP/1.1, HTTP/2, and HTTP/3 across 4 architectural parameters. (8 Marks)

1. **Transport Layer:** HTTP/1.1 and HTTP/2 run over standard TCP; HTTP/3 runs over QUIC (UDP-based).
2. **Multiplexing:** HTTP/1.1 uses sequential request-response pipelining (suffers from Head-of-Line blocking); HTTP/2 uses binary frame multiplexing over a single TCP connection; HTTP/3 achieves stream-level multiplexing without TCP Head-of-Line blocking.
3. **Header Compression:** HTTP/1.1 uses plain text headers without compression; HTTP/2 uses HPACK; HTTP/3 uses QPACK.
4. **Handshake Latency:** HTTP/1.1 & HTTP/2 require 2–3 RTTs (TCP + TLS); HTTP/3 achieves 0–RTT connection resumption.